

CRITICAL CLEAN AIR TREATMENT • WHD SERIES

Heatless Twin-Tower Desiccant Air Dryers

Ultra-Deep Moisture Adsorption Delivering Guaranteed -40°C to -70°C Pressure Dew Point

Engineered for Sub-Zero -40°C Pressure Dew Point (ISO 8573-1 Class 2): Completely prevents line freeze-ups, pipeline corrosion, and moisture condensation in pharmaceutical cleanrooms, robotic paint booths, laser cutting heads, and electronic assembly lines.

The Win Equipments WHD Series utilizes dual vertical pressure vessels filled with premium spherical activated alumina desiccant. Operating on a precision 8-minute Pressure Swing Adsorption (PSA) cycle, one tower adsorbs moisture under pressure while the opposing tower regenerates using an expanded dry air purge stream (12–15%).

PSA HEATLESS CYCLE

-40°C TO -70°C DEW POINT

ACTIVATED ALUMINA MEDIA

4.0 TO 16.0 BAR G



300–2000

CAPACITY (CFM)

-40°C

STANDARD DEW POINT

16 bar g

RATED PRESSURE

< 0.25 bar

BED PRESSURE DROP

CORE ENGINEERING ADVANTAGES

High Crush Strength Media

Spherical 4–8mm activated alumina desiccant with >20% dynamic water retention capacity by weight, high mechanical crush resistance, and zero dusting to protect downstream tools.

PLC Precision Cycling

Microprocessor / Omron solid-state cycle timer controls fast-acting pneumatic angle-seat switching valves rated for >1,000,000 continuous switching cycles.

Low-Noise Blowdown

Dual heavy-duty exhaust purge mufflers attenuate depressurization acoustics below 75 dBA, ensuring full compliance with factory floor noise regulations.

TARGET INDUSTRIAL APPLICATIONS

Pharmaceutical API & Sterile Processing

Semiconductor & Electronics Assembly

Automotive Clear-Coat Paint Lines

CNC Fiber Laser Cutting Assist Gas

Breweries & Beverage Packaging

Cryogenic & Chemical Plants

TECHNICAL SPECIFICATIONS & ENGINEERING MATRIX

Standard design conditions: Operating pressure 7.0 bar g (max 16.0 bar g), Inlet air temperature 35°C to 45°C, Outlet pressure dew point -40°C (optional -70°C).

MODEL CODE	CFM	Flow (m ³ /hr)	Rated bar	Connection	Voltage	Dimensions (mm)	Weight (kg)
WHD - 030	300 CFM	510 m ³ /h	7-16 bar	1½" NB FFlange	220V / 1Ph / 50Hz	1450 × 1100 × 2500	450 kg
WHD - 040	400 CFM	680 m ³ /h	7-16 bar	2" NB FFlange	220V / 1Ph / 50Hz	1450 × 1200 × 2500	630 kg
WHD - 050	500 CFM	850 m ³ /h	7-16 bar	3" NB FFlange	220V / 1Ph / 50Hz	1650 × 1250 × 2300	720 kg
WHD - 060	600 CFM	1020 m ³ /h	7-16 bar	3" NB FFlange	220V / 1Ph / 50Hz	1650 × 1250 × 2500	910 kg
WHD - 075	750 CFM	1275 m ³ /h	7-16 bar	3" NB FFlange	220V / 1Ph / 50Hz	1800 × 1300 × 2600	1120 kg
WHD - 100	1000 CFM	1700 m ³ /h	7-16 bar	4" NB FFlange	220V / 1Ph / 50Hz	2000 × 1400 × 2700	1450 kg
WHD - 150	1500 CFM	2550 m ³ /h	7-16 bar	5" NB FFlange	220V / 1Ph / 50Hz	2300 × 1600 × 2900	2100 kg
WHD - 200	2000 CFM	3400 m ³ /h	7-16 bar	6" NB FFlange	220V / 1Ph / 50Hz	2600 × 1800 × 3100	2850 kg

STANDARD PRE & AFTER FILTRATION PACKAGE

To guarantee -40°C performance and safeguard desiccant longevity, Win Equipments supplies pre-piped upstream coalescing pre-filters (0.01 micron / 0.01 mg/m³ oil carryover) and downstream particulate after-filters (1.0 micron) with differential pressure indicators.

Dew Point Dependent Switching (DDS) Option: Optional online capacitive ceramic dew point sensor extends drying cycles when inlet air is lightly loaded, reducing purge air consumption from 15% down to under 5% during low pneumatic demand periods.

Request Technical Submittal & Commercial Quotation

Direct factory engineering consultation for custom capacity sizing, P&ID schematics & dimensional drawings.

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